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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

SUPERMARKET CHECKOUT AND DISCOUNT BILLING SYSTEM

Submitted by

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Semester: **II**

DECLARATION

I hereby declare that the project entitled "**Supermarket Checkout And Discount Billing System"** submitted in partial fulfillment of the requirements for the course **B.Sc Information Technology** is a **record of my original work** carried out under the guidance of **Mugilan R**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

Lohitasree SR

CERTIFICATE

This is to certify that the project entitled **"Supermarket Checkout And Discount Billing System"** is a **bona fide work** carried out by **Lohitasree SR ,B.Sc Information Technology**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming.**

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

_____

Head of the Department
(Technical Head, KGCAS)

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ABSTRACT

The Supermarket Checkout and Discount Billing System is a Java-based application designed to automate the billing process in supermarkets. In many small retail stores, billing operations are performed manually, which can lead to calculation errors, delays at checkout counters, and inefficient record management. This project aims to simplify the checkout process by providing an automated system that calculates the total cost of purchased items and generates an accurate bill.

The system allows users to select products from a predefined list, enter the quantity of each item, and automatically calculate the total bill including applicable discounts and taxes such as GST. The application is developed using Core Java programming concepts, including classes, objects, methods, conditional statements, loops, and exception handling. The project can be implemented either as a console-based application or using Java Swing for a graphical user interface.

The main objective of the system is to improve billing efficiency, reduce human errors, and provide faster service to customers at the checkout counter. The system generates a clear and structured bill that displays purchased items, quantities, prices, subtotal, discount applied, tax, and the final payable amount.

This project demonstrates the practical implementation of object-oriented programming concepts in Java and provides a basic foundation for developing more advanced retail management systems in the future.

1. INTRODUCTION

In the retail industry, efficient billing systems play a crucial role in managing supermarket operations and ensuring customer satisfaction. Traditionally, many small supermarkets rely on manual billing methods, where staff calculate the total bill using handwritten notes or simple calculators. This manual process is time-consuming and can often lead to calculation errors, delays at checkout counters, and difficulty in maintaining proper billing records.

With the advancement of computer technology, automated billing systems have become essential for improving efficiency and accuracy in retail management. A computerized checkout system helps supermarket staff quickly generate bills, calculate totals automatically, apply discounts, and maintain organized records of products and transactions.

The Supermarket Checkout and Discount Billing System is a Java-based application designed to automate the process of product billing and invoice generation. The system allows the user to select items from a predefined product list, enter the quantity of each item, and automatically calculate the total bill including applicable discounts and taxes. The application generates a clear and structured invoice displaying purchased items, quantities, prices, subtotal, discount, tax, and the final payable amount.

This project is developed using Core Java programming concepts such as classes, objects, methods, conditional statements, loops, and exception handling. The application can be implemented either as a console-based program or using Java Swing for a graphical user interface (GUI).

The primary goal of this system is to reduce manual effort, improve billing accuracy, and provide faster service to customers at the checkout counter. This project also demonstrates how Java programming can be applied to develop simple real-world retail applications.

1.1 Objective

The main objective of the Supermarket Checkout and Discount Billing System is to develop a simple and efficient billing application using Core Java that automates the process of product checkout and discount-based bill generation in supermarkets.

The specific objectives of this project are:

->To design and develop a Java-based supermarket billing system.

->To automate the calculation of product prices, discounts, and total bill.

->To reduce manual errors in billing calculations.

->To provide a user-friendly system for supermarket staff to generate bills quickly.

->To demonstrate the implementation of Core Java programming concepts such as classes, objects, methods, loops, and conditional statements.

->To generate a clear and organized bill for customers.

1.2 Overview

The Supermarket Checkout and Discount Billing System is a menu-driven application developed using the Java programming language. The system provides a digital platform where supermarket staff can select products from a predefined list and generate bills automatically at the checkout counter.

The system displays a list of products along with their prices. The user can select the required items and enter the quantity purchased by the customer. The program calculates the total price for each item, applies applicable discounts, and generates a final bill including taxes such as GST.

The system helps simplify the checkout process by eliminating manual calculations and reducing the chances of billing errors. The application can be developed as a console-based program or a graphical user interface using Java Swing.

This project demonstrates the practical use of Java programming in developing real-world applications for retail environments such as supermarkets and stores.

1.3 Features

The Supermarket Checkout and Discount Billing System includes the following main features:

1.Product Display – Displays a list of available products in the supermarket along with their prices.

2.Item Selection – Allows the user to select products and enter the quantity purchased by the customer.

3.Automatic Bill Calculation – Automatically calculates the total amount based on selected items and quantities.

4.Discount Calculation – Applies discounts based on predefined rules or total purchase amount.

5.Tax Calculation – Calculates applicable taxes such as GST and adds them to the total bill.

6.Invoice Generation – Generates a detailed bill showing product names, quantities, prices, subtotal, discount, tax, and final amount.

7.User-Friendly Interface – Provides a simple and easy-to-use menu-driven interface.

8.Fast Checkout Process – Reduces the time required to generate customer bills and improves efficiency at the billing counter

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in the existing system and proposing a computerized solution that improves efficiency and accuracy.

2.1 Problem Definition

In many small supermarkets and retail stores, billing is often performed manually using paper or simple calculators. This traditional method creates several challenges.

Some common problems include:

1.Errors in bill calculation

2.Time-consuming checkout process

3.Difficulty in managing multiple items and customers

4.Lack of proper billing records

5.Inefficient service during peak hours

Manual billing systems also increase the workload for staff and can lead to customer dissatisfaction due to delays or incorrect billing amounts.

Therefore, there is a need for an automated billing system that can quickly calculate the total bill, apply discounts, manage product details, and generate accurate invoices.

The Supermarket Checkout and Discount Billing System is designed to solve these problems by providing a computerized solution that automates the checkout and billing process.

2.2 Project Overview

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2.3 Scopes and Limitations

Scope

The scope of the Supermarket Checkout and Discount Billing System includes the following:

1. Managing supermarket product items
2. Processing customer purchases at checkout
3. Calculating the total bill automatically
4. Applying discounts and taxes
5. Generating a detailed invoice for the customer
6. Demonstrating the practical use of Java programming concepts

The system can be used in:

1. Supermarkets
2. Retail stores
3. Grocery shops
4. Mini-marts

5.Small-scale businesses

Limitations

1.Limited to predefined product lists (no dynamic database integration)

2.Does not support online payment systems

3.Suitable mainly for small and medium-scale supermarkets

4.Limited security and user authentication features

5.No inventory management (stock update not included)

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The Supermarket Checkout and Discount Billing System is designed as a menu-driven application where users can select products, enter quantities, apply discounts, and generate the final bill automatically.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

3.1 Modules

The main modules of the **Supermarket Checkout And Discount Billing System** are explained below:

1. Menu Display Module

This module is responsible for displaying the list of available products along with their prices. It helps the user to select the required items for billing.

Example Product List:

1.Rice (1kg) – ₹60

2.Milk (1L) – ₹50

3.Bread – ₹40

4.Soap – ₹30

5.Cooking Oil (1L) – ₹150

The product list is displayed when the program starts so that the user can easily view available items.

2. Item Selection Module

This module allows the user to select products from the list and enter the required quantity.

Functions of this module include:

1.Accepting user input for product selection

2.Accepting quantity of selected products

3.Storing selected items temporarily

4.Allowing multiple product selection

The system enables users to select multiple items before proceeding to billing.

3. Billing and discount Calculation Module

This module performs all billing and discount-related calculations.

The system calculates:

Total price of each product

Subtotal of all selected items

Applicable discount (e.g., percentage discount, seasonal offers, bulk purchase discount)

GST or tax amount

Final payable amount

All calculations are performed automatically using Java programming logic.

Example Calculation:

Item Total = Price × Quantity

Subtotal = Sum of all item totals

Discount = (Discount % × Subtotal)

GST = 5% of (Subtotal − Discount)

Final Bill = Subtotal − Discount + GST

4. Bill Generation Module

This module generates the final bill for the customer after applying discounts and taxes.

The bill includes:

Product name

Quantity

Price

Subtotal

Discount amount

GST amount

Final bill amount -

----- SUPERMARKET BILL ------

Item	Qty	Price
Rice (1kg)	2	120
Milk (1L)	2	100
Soap	1	30

Subtotal	250
Discount (10%)	25
GST (5%)	11.25
Total	236.25

Thank You! Visit Again!

5. Exit Module

This module allows the user to exit the application safely after completing the billing process.

When the user selects the exit option, the system terminates the program.

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the Supermarket Checkout and Discount Billing System, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

1.Problem Analysis

2.System Design

3.Program Development

4.Testing and Validation

5.Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of supermarket operations.

The main problems identified in manual billing systems are:

1.Manual calculations may lead to billing errors

2.Preparing bills manually takes more time

3.Difficulty in handling multiple customers at checkout

4.Lack of automated records for transactions

5.No proper system for applying discounts accurately

Based on these problems, the system requirements were defined to develop an automated billing application that simplifies supermarket checkout operations and ensures accurate billing with discounts.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

1.Product Display Module

2.Item Selection Module

3.Discount Calculation Module

4.Bill Calculation Module

5.Invoice Generation Module

6.Exit Module

The program flow and structure are designed so that users can easily interact with the system and perform billing operations efficiently.

4.3. Program Development

After designing the system, the program is implemented using the Java programming language.

The following steps were followed during development:

1.Creation of classes and methods

2.Implementation of menu-driven logic

3.Writing code for product selection and billing

4.Implementing discount calculation logic

5.Handling user input

Displaying the final bill with discount and tax details

4.4. Testing and Validation

Testing ensures that the supermarket billing program works correctly and produces accurate results.

Different test cases were used to verify the functionality of item selection, discount calculation, and final billing.

Example Test Cases:

Test Case	Expected Result
Select product	Product details displayed correctly
Enter quantity	Correct quantity accepted
Apply discount	Discount applied correctly
Calculate bill	Accurate total displayed
Invalid input	Error handled properly

This phase ensures that the system runs smoothly without errors and generates correct billing results.

4.5. Result and Documentation

After successful testing, the results of the system are documented.

The system generates the final bill based on selected products, quantities, and applicable discounts.

The results are demonstrated through program execution and screenshots showing input and output.

Proper documentation is prepared to explain the design, implementation, and working of the supermarket checkout system.

4.6. Algorithm

Algorithm for the Supermarket Checkout and Discount Billing System:

1.Start the program.

2.Display the list of products with prices.

3.Ask the user to select a product.

4.Enter the quantity of the selected product.

5.Calculate the cost of the selected item.

6.Add the cost to the total bill.

7.Repeat steps 3–6 for additional products if required.

8.Check for applicable discounts (e.g., bulk discount or special offers).

9.Apply the discount to the total amount.

10.Calculate GST or tax amount.

11.Calculate the final bill amount.

12.Display the final bill to the user.

13.End the program.

4.7. Flowchart

Students should draw a flowchart showing the program flow of the Supermarket Checkout and Discount Billing System.

Example flow:

Start

↓

Display Product List

↓

Select Product

↓

Enter Quantity

↓

Calculate Item Cost

↓

Add to Total Bill

↓

More Items? (Yes/No)

↓

If Yes → Repeat Selection Process

↓

If No → Apply Discount

↓

Calculate GST/Tax

↓

Calculate Final Bill

↓

Display Invoice/Bill

↓

End

Students can draw using:

- draw.io
- Lucidchart
- MS Visio

5. SOURCE CODE

```
import java.util.Scanner;

class Product {
    String name;
    double price;
    int quantity;

    Product(String name, double price, int quantity) {
        this.name = name;
        this.price = price;
        this.quantity = quantity;
    }

    double getTotal() {
        return price * quantity;
    }
}

public class SupermarketBillingSystem {

    public static double calculateDiscount(double total) {
        double discount = 0;
```

```
    if (total >= 5000) {
        discount = total * 0.20; // 20% discount
    } else if (total >= 3000) {
        discount = total * 0.15; // 15% discount
    } else if (total >= 1000) {
        discount = total * 0.10; // 10% discount
    }

    return discount;
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.println("===== Supermarket Checkout System =====");
    System.out.print("Enter number of items: ");
    int n = sc.nextInt();

    Product[] items = new Product[n];
    double grandTotal = 0;

    for (int i = 0; i < n; i++) {
        System.out.println("\nEnter details for item " + (i + 1));

        System.out.print("Product Name: ");
        sc.nextLine(); // clear buffer
        String name = sc.nextLine();

        System.out.print("Price: ");
```

```
        double price = sc.nextDouble();

        System.out.print("Quantity: ");
        int quantity = sc.nextInt();

        items[i] = new Product(name, price, quantity);
        grandTotal += items[i].getTotal();
    }

    double discount = calculateDiscount(grandTotal);
    double finalAmount = grandTotal - discount;

    // Bill printing
    System.out.println("\n========= BILL =========");
    System.out.printf("%-15s %-10s %-10s %-10s\n", "Item", "Price", "Qty", "Total");

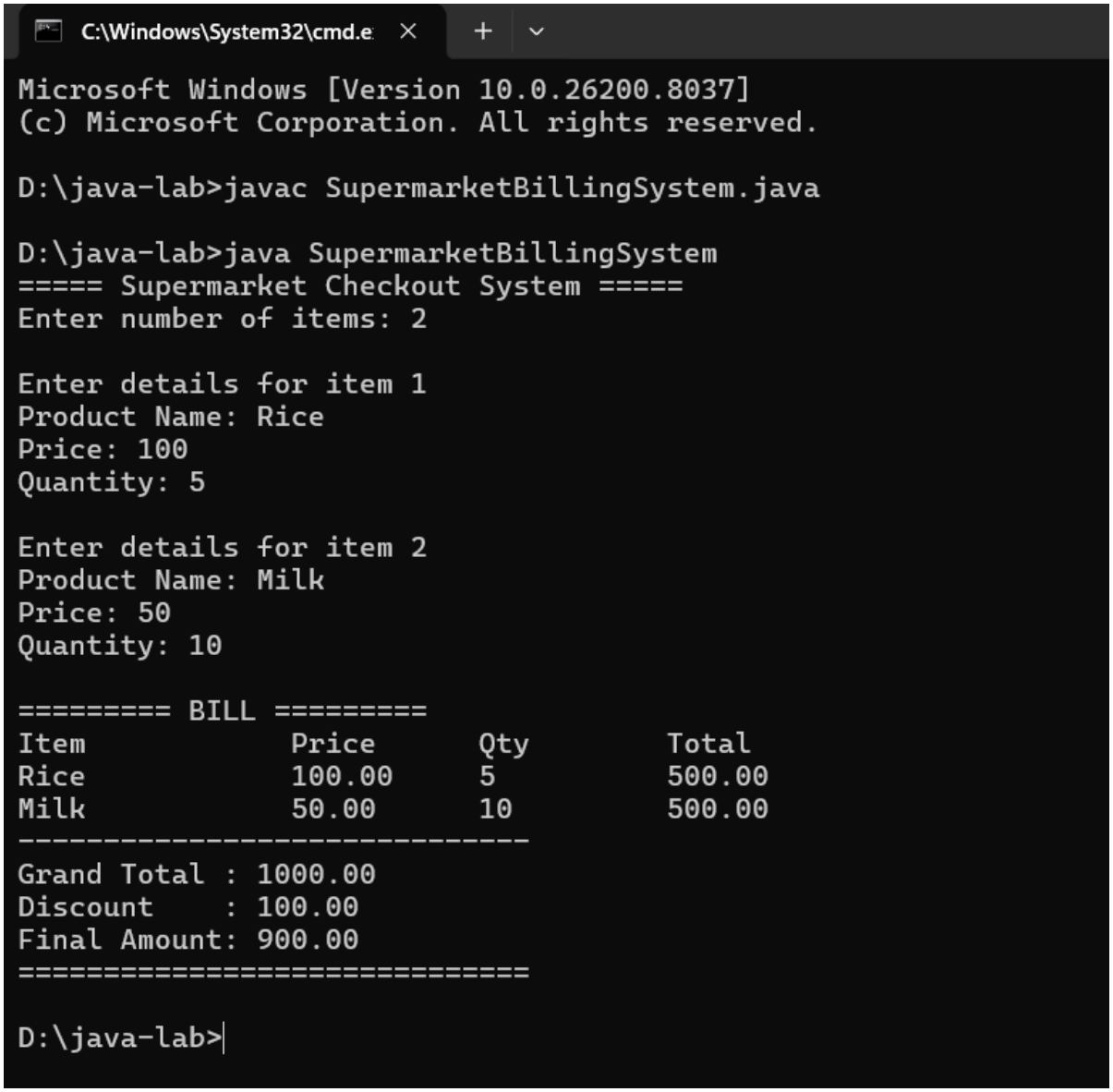
    for (Product p : items) {
        System.out.printf("%-15s %-10.2f %-10d %-10.2f\n",
                p.name, p.price, p.quantity, p.getTotal());
    }

    System.out.println("-----------------------------");
    System.out.printf("Grand Total : %.2f\n", grandTotal);
    System.out.printf("Discount    : %.2f\n", discount);
    System.out.printf("Final Amount: %.2f\n", finalAmount);
    System.out.println("=============================");

    sc.close();
}
```

}

6. INPUT AND OUTPUT SCREENS


C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.
D:\java-lab>javac SupermarketBillingSystem.java
D:\java-lab>java SupermarketBillingSystem
===== Supermarket Checkout System =====
Enter number of items: 2
Enter details for item 1
Product Name: Rice
Price: 100
Quantity: 5
Enter details for item 2
Product Name: Milk
Price: 50
Quantity: 10
========= BILL =========
Item Price Qty Total
Rice 100.00 5 500.00
Milk 50.00 10 500.00
Grand Total : 1000.00
Discount : 100.00
Final Amount: 900.00
D:\java-lab>

DISCUSSION & CONCLUSION

7.1 Discussion

The Supermarket Checkout and Discount Billing System was developed to automate the billing process in supermarkets. The system allows users to select products, enter quantities, apply discounts, and generate bills efficiently.

During the development process, Core Java programming concepts such as loops, conditional statements, methods, and object-oriented principles were applied. The system successfully performs product-wise billing, applies discounts based on conditions (such as offers or quantity), and generates accurate invoices.

The system was tested with different inputs to ensure that it performs all calculations correctly, including discount and tax computation. The results show that the system reduces manual effort, minimizes billing errors, and improves checkout speed and efficiency.

7.2 Conclusion

The Supermarket Checkout and Discount Billing System successfully demonstrates the practical application of Core Java programming concepts in developing a real-world business application.

The system automates the checkout process, reduces calculation errors, and provides faster service to customers. It simplifies supermarket billing operations by accurately calculating totals, applying discounts, and generating clear invoices.

This project helps students understand how Java programming can be used to build practical software systems. The system can be further enhanced by adding features such as barcode scanning, database connectivity, graphical user interfaces, and digital payment options.

7. **REFERENCES**

The following resources were used during the development of this project:

Books

• Herbert Schildt – Java: The Complete Reference

• E. Balagurusamy – Programming with java

Websites

https://www.geeksforgeeks.org/

https://www.javatpoint.com/

https://www.w3schools.com/